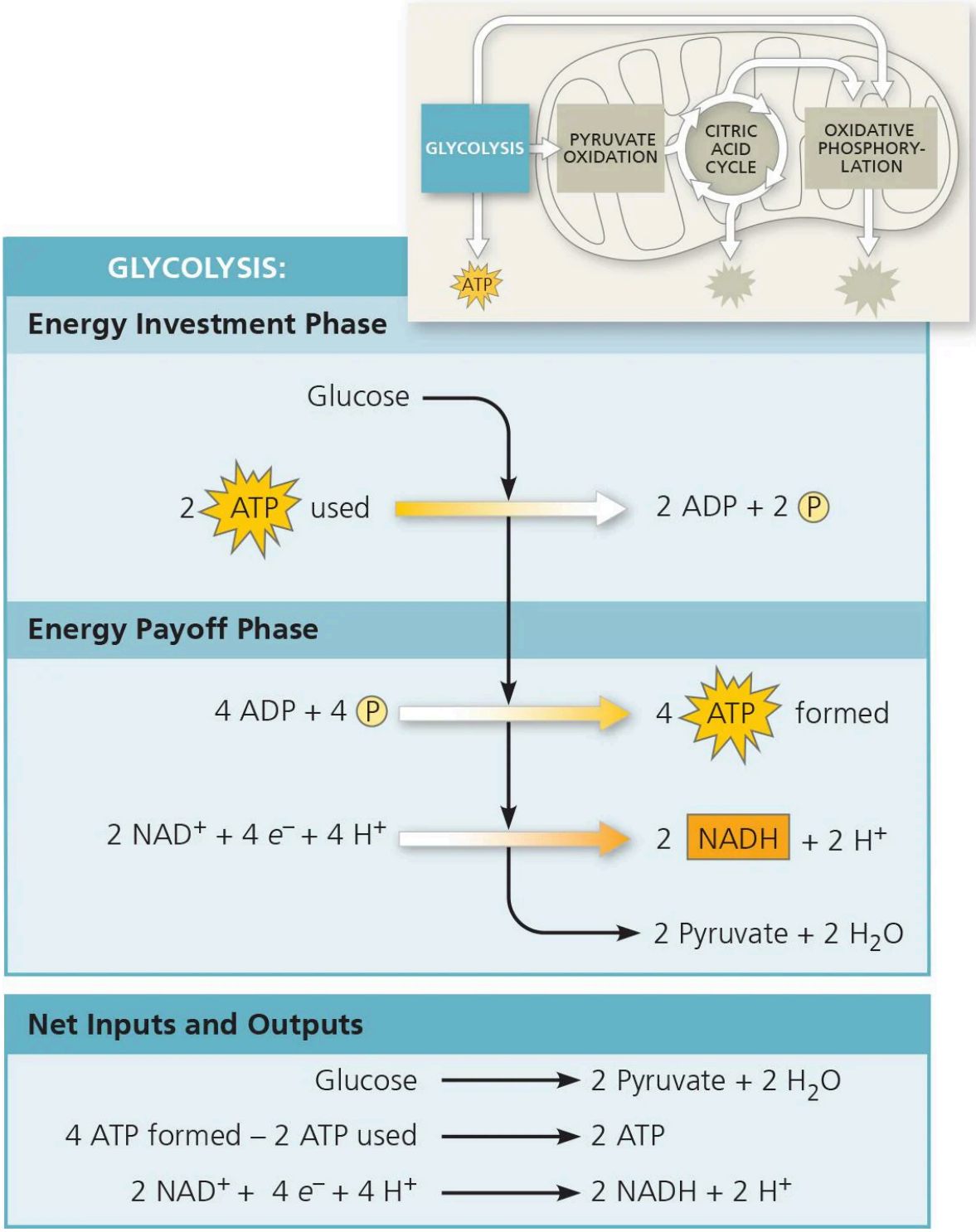
# 9.2: Glycolysis harvests chemical energy by oxidizing glucose to pyruvate

The word *glycolysis* means “sugar splitting,” and that is exactly what happens during this pathway. Glucose, a six-carbon sugar, is split into two three-carbon sugars. These smaller sugars are then oxidized and their remaining atoms rearranged to form two molecules of pyruvate. (Pyruvate is the ionized form of pyruvic acid.)

As summarized in **Figure 9.7**, glycolysis can be divided into two phases: the energy investment phase and the energy payoff phase. During the energy investment phase, the cell actually spends ATP. This investment is repaid with interest during the energy payoff phase, when ATP is produced by substrate-level phosphorylation and NAD+ is reduced to NADH by electrons released from the oxidation of glucose. The net energy yield from glycolysis, per glucose molecule, is 2 ATP plus 2 NADH. The ten steps of the glycolytic pathway are shown in **Figure 9.8**.

**Figure 9.7**

The inputs and outputs of glycolysis.

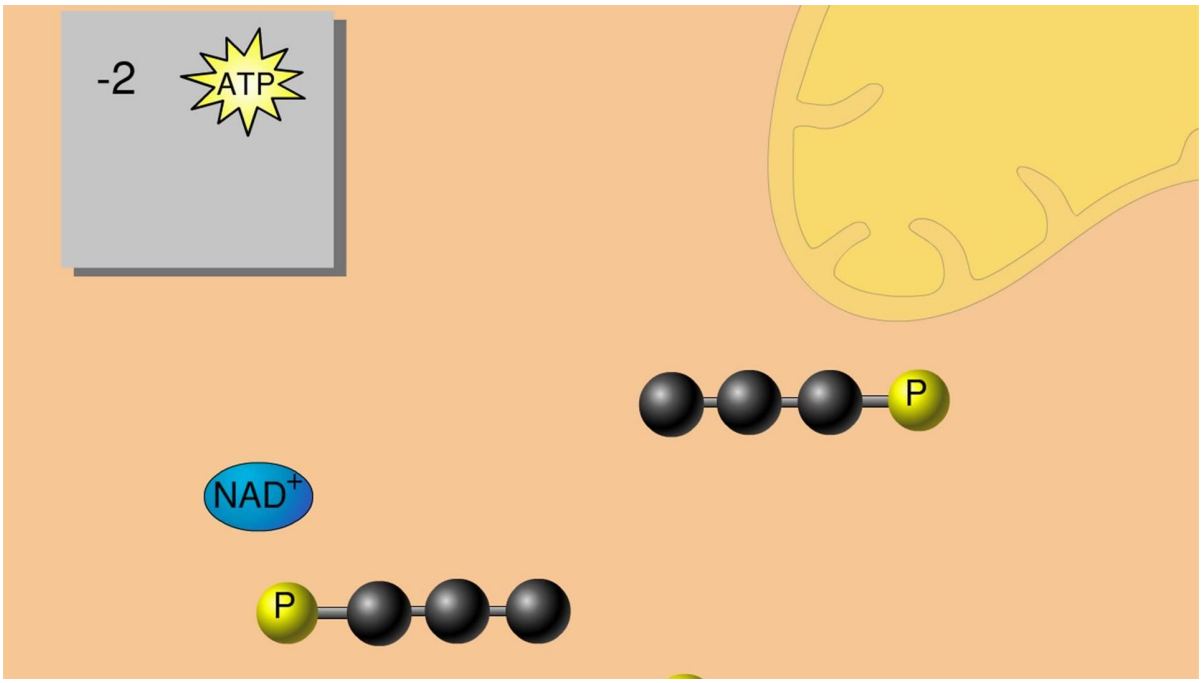



Overview: Glycolysis takes place in the cytoplasm and produces a small amount of ATP. The outputs from glycolysis will enter pyruvate oxidation, the citric acid cycle, and oxidative phosphorylation inside the mitochondrion. Inputs and output from glycolysis are summarized as follows: Energy Investment Phase: 2 A T P goes to 2 A D P plus 2 inorganic phosphates. Glucose enters the pathway. Energy payoff phase: 4 A D P plus 4 inorganic phosphate goes to 4 A T P. 2 N A D plus, plus 4 e minus plus 4 H plus goes to 2 N A D H plus

2 H plus. Glucose is split into 2 pyruvates plus 2 H 2 0. Net: Glucose goes to 2 pyruvates plus 2 H 2 0. 4 ATP formed minus 2 A T P used equals to 2 A T P. 2 N A D plus, plus 4 e minus plus 4 H plus goes to 2 N A D H plus 2 H plus.

**Animation**

Glycolysis

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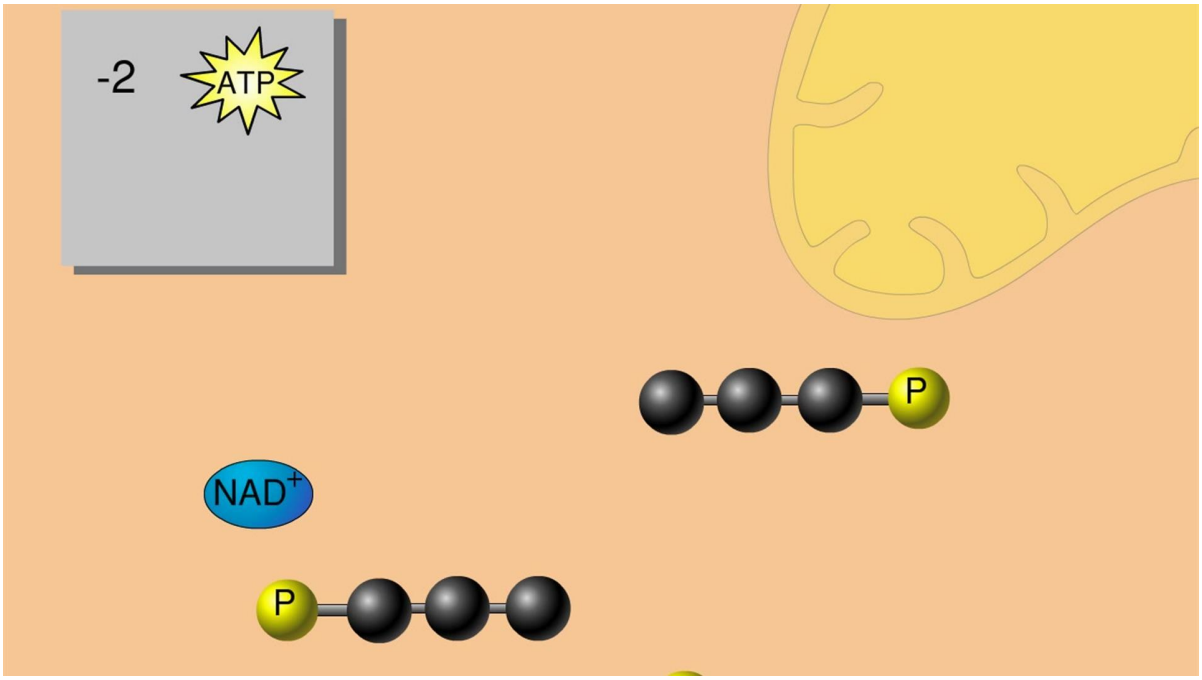

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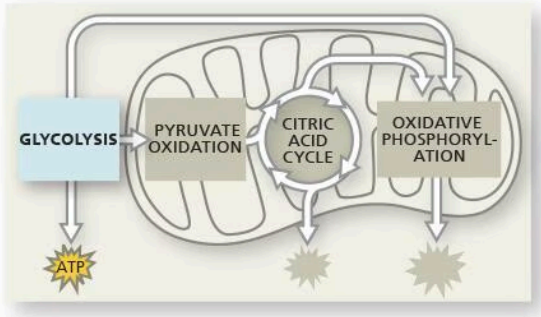
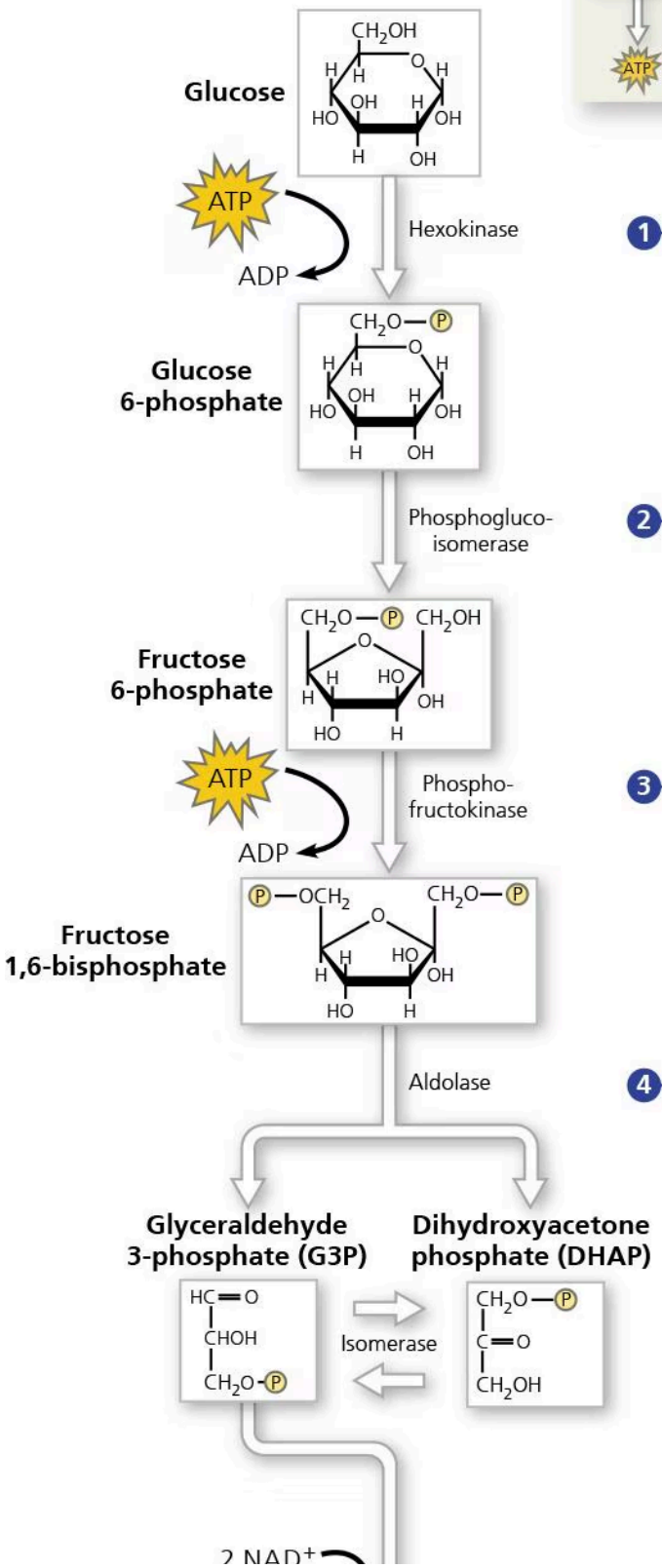
**Figure 9.8**

The steps of glycolysis.

## GLYCOLYSIS: Energy Investment Phase

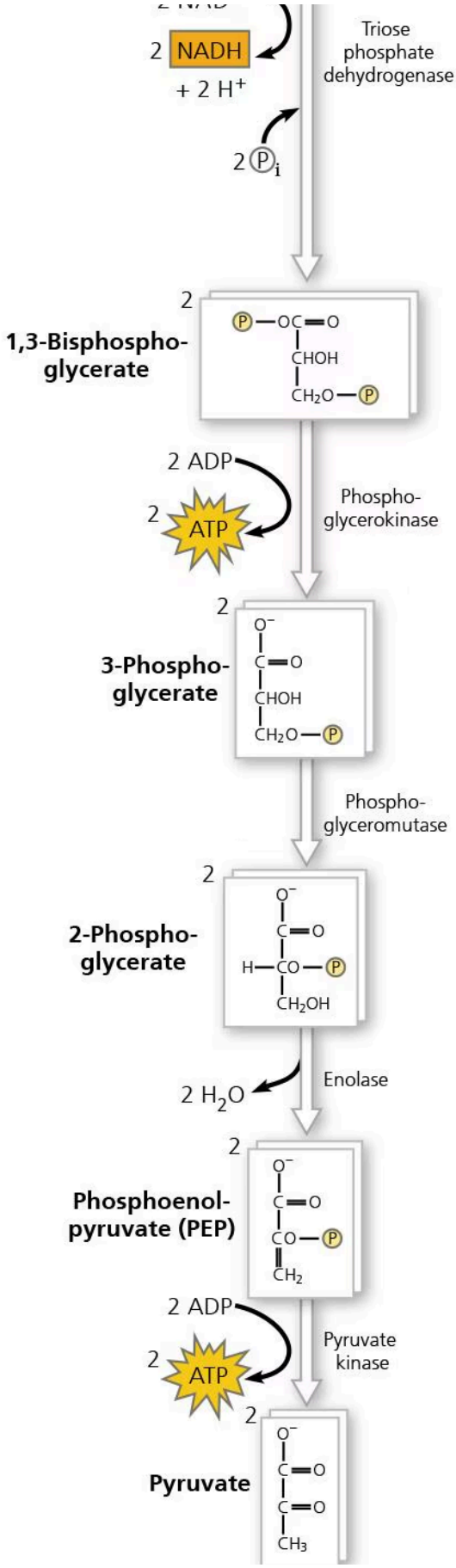



1. Hexokinase transfers a phosphate group from ATP to glucose, making it more chemically reactive. The charge on the phosphate also traps the sugar in the cell.
2. Glucose 6-phosphate is converted to fructose 6-phosphate.
3. Phosphofructokinase transfers a phosphate group from ATP to the opposite end of the sugar, investing a second molecule of ATP. This is a key step for regulation of glycolysis.
4. Aldolase cleaves the sugar molecule into two different three-carbon sugars.
5. Conversion between DHAP and G3P: This reaction never reaches equilibrium; G3P is used in the next step as fast as it forms.

The energy payoff phase occurs after glucose is split into two three-carbon sugars. Thus, the coefficient 2 precedes all molecules in this phase.

GLYCOLY




6 Two sequential reactions: (1) G3P is oxidized by the transfer of electrons to $NAD^+$, forming NADH. (2) Using energy from this exergonic redox reaction, a phosphate group is attached to the oxidized substrate, making a high-energy product.

7 The phosphate group is transferred to ADP (substrate-level phosphorylation) in an exergonic reaction. The carbonyl group of G3P has been oxidized to the carboxyl group ($—COO^-$) of an organic acid (3-phosphoglycerate).

8 This enzyme relocates the remaining phosphate group.

9 Enolase causes a double bond to form in the substrate by extracting a water molecule, yielding phosphoenolpyruvate (PEP), a compound with a very high potential energy.

10 The phosphate group is transferred from PEP to ADP (a second example of substrate-level phosphorylation), forming pyruvate.

SIS: Energy Payoff Phase

Hexokinase transfers a phosphate group from ATP to glucose, making it more chemically reactive. The charge on the phosphate also traps the sugar in the cell. The structure of glucose is a hexagon (a six-carbon ring with carbons 1-5 forming 5 corners of the ring and O attached to carbon 5 at the sixth corner). O H and H are attached to carbons 1-4. C H 2 O H is attached to carbon 5, as is O in the sixth corner of the ring. One A T P is used (going to A D P) with the enzyme hexokinase to form glucose 6-phosphate, which has the same structure as glucose, except that there is a phosphate group (from A T P) attached above carbon 5, forming C H 2 O attached to a phosphate group. Glucose 6-phosphate is converted to fructose 6-phosphate. The enzyme involved in this conversion is phosphoglucoisomerase. Glucose 6-phosphate (structure described in a) is converted to fructose 6-phosphate with the enzyme phosphoglucoisomerase. The structure of fructose 6-phosphate is a pentagon, with carbon at all unlabeled corners. Attached to carbon 1 are C H 2 O H and H. Attached to carbons 2 and 3 are O H and H. Attached to carbon 4 are H and C H 2 O H attached to a phosphate group. At the point of the pentagon is an O. Phosphofructokinase transfers a phosphate group from A T P to the opposite end of the sugar, investing a second molecule of A T P. This is a key step for the regulation of glycolysis. Fructose 6-phosphate is converted to fructose 1,6-bisphosphate as the enzyme phosphofructokinase transfers a phosphate group from ATP. Fructose 1,6-bisphosphate has the same structure as fructose 6 phosphate (described in b), with the addition of a phosphate group (from A T P) attached above carbon 1, forming C H 2 O attached to a phosphate group. Aldolase cleaves the sugar molecule into two different three-carbon sugars. Aldolase cleaves fructose 1,6-bisphosphate (structure described in c) into glyceraldehyde 3-phosphate (G3P) and dihydroxyacetone phosphate (D H A P). Structure of glyceraldehyde 3-phosphate is a 3-carbon chain with an H and an O (double bond) attached to carbon 1, an H O H attached to carbon 2 and an H 2 0, single bond, P attached to carbon 3. Structure of dihydroxyacetone phosphate is a 3 carbon chain with H 2 0 attached to a phosphate group attached to carbon 1, an O (double bond) attached to carbon 2 and an H 2 O H attached to carbon 3. Conversion between D H A P and G 3 P: This reaction never reaches equilibrium; G 3 P is used in the next step as fast as it forms. Isomerase catalyzes the reversible reaction between glyceraldehyde 3-phosphate and dihydroxyacetone phosphate (structures in d). Continued on next page.

Two sequential reactions: (1) 2 N A D plus are reduced to 2 N A D H plus 2 H plus. (2) G 3 P is converted to 1,3 bisphosphoglycerate.

Two sequential reactions: (1) G3P is oxidized by the transfer of electrons to N A D plus, forming N A D H. (2) Using energy from this exergonic redox reaction, a phosphate group is attached to the oxidized substrate, making a high-energy product.

In a reaction catalyzed by triose phosphate dehydrogenase and with an input of 2 inorganic phosphate molecules, glyceraldehyde 3-phosphate (G 3 P) is converted to 1,3 bisphosphoglycerate. The structure is a 3-carbon chain P, single bond, O and O (double bond) attached to carbon 1, H O H attached to carbon 2, and H 2 0 attached to a phosphate group attached to carbon 3. The phosphate group is transferred to ADP

(substrate-level phosphorylation) in an exergonic reaction. The carbonyl group of G3P has been oxidized to the carboxyl group (C O O minus) of an organic acid (3-phosphoglycerate). 2 A T P are formed from 2 A D P through substrate-level phosphorylation. In a reaction catalyzed by phosphoglycerokinase, 1,3-Bisphosphoglycerate (structure described in f) is converted to 3-phosphoglycerate. Structure is a 3 carbon chain with an O minus and another O (double bond) attached to carbon 1, an H O H attached to carbon 2, and an H 2 0 attached to a phosphate group attached to carbon 3. This enzyme relocates the remaining phosphate group. 3-phosphoglycerate is converted to 2-phosphoglycerate in a reaction catalyzed by phosphoglyceromutase. 3-phosphoglycerate (structure described in g) is converted to 2-phosphoglycerate. Structure is the same as 3-phosphoglycerate, except the enzyme phosphoglyceromutase relocates the phosphate group from the location in 3-phosphoglycerate to the second carbon in 2-phosphoglycerate so that H and O with a phosphate group are now attached at carbon 2. Enolase causes a double bond to form in the substrate by extracting a water molecule, yielding phosphoenolpyruvate (P E P), a compound with a very high potential energy. 2-phosphoglycerate is converted to P E P when enolase extracts a water molecule, causing a double bond to form in P E P. Structure is the same as 2-phosphoglycerate, except there is a double bond between carbon 3 and C H 2 (was C H 2 O H) and there is no longer an H on carbon 2. The phosphate group is transferred from P E P to A D P (a second example of substrate-level phosphorylation), forming pyruvate. PEP (structure described in i) is converted to pyruvate in a reaction catalyzed by pyruvate kinase. 2 A T P are formed from 2 A D P during the conversion by substrate level phosphorylation. Structure of pyruvate is the same as P E P, except the phosphate group has been removed from carbon 2 and carbon 3 is now C H 3.

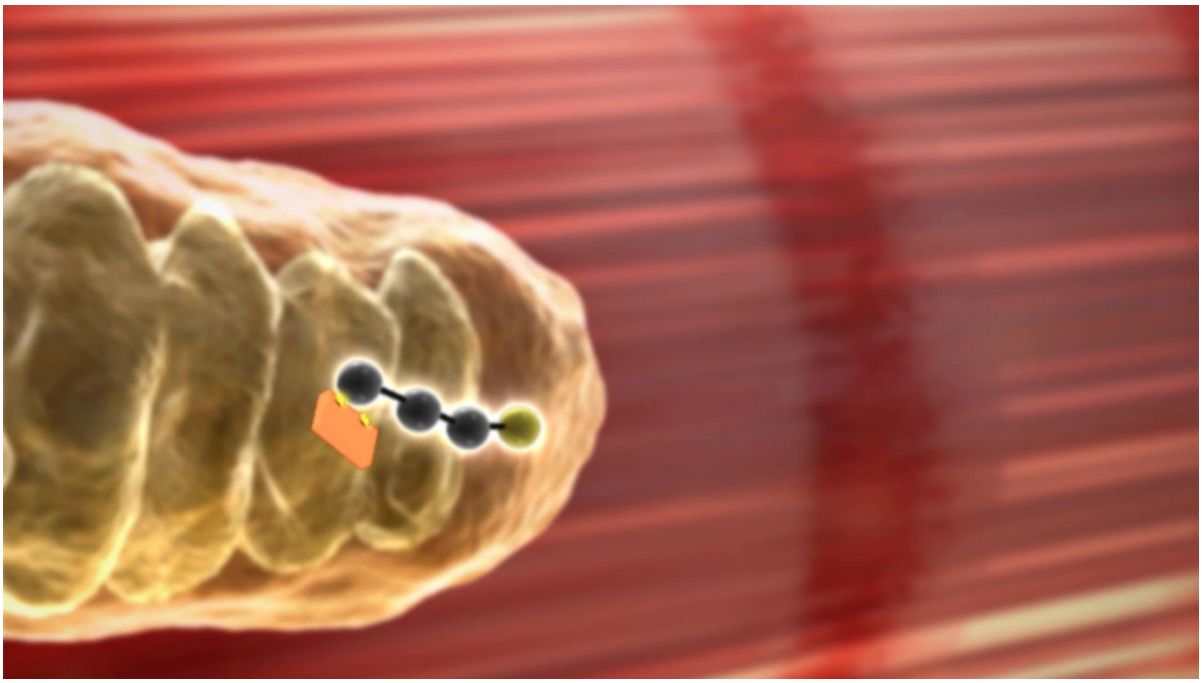
Glycolysis, a source of ATP and NADH, takes place in the cytosol. Two of the enzymes (in **STEPS 1** and **3**) are shown in Figure 6.32b.

**WHAT IF?** *What would happen if you removed the dihydroxyacetone phosphate generated in step 4 as fast as it was produced?*

*For suggested answer, see*Appendix A*.*

**BioFlix® Animation**

Glycolysis



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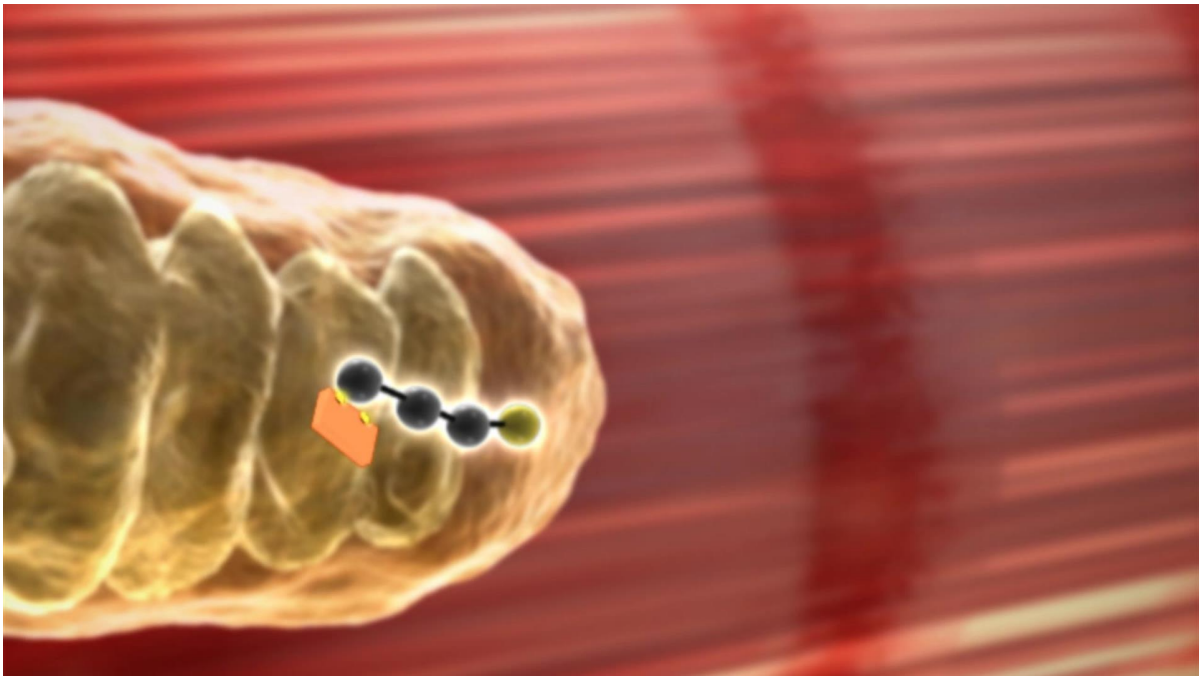

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All of the carbon originally present in glucose is accounted for in the two molecules of pyruvate; no carbon is released as $CO_2$ during glycolysis. Glycolysis occurs whether or not $O_2$ is present. However, if $O_2$ *is* present, the chemical energy stored in pyruvate and NADH can be extracted by pyruvate oxidation, the citric acid

cycle, and oxidative phosphorylation.